

Cahier des Charges

Comparative Study of Data Augmentation Effects on 3D Mesh Classification Models

1. Project Context

Several deep learning architectures have been proposed for mesh-based classification. However, the influence of geometric data augmentation on these architectures is not always clearly studied, especially across different datasets and representations.

The objective of this project is to perform a rigorous experimental comparison of several state-of-the-art architectures using:

- original datasets
- augmented datasets

The study aims to evaluate the impact of augmentation on:

- classification performance
- generalization capability
- training stability
- computational cost

2. Objectives

The project consists of:

1. Using the official implementations of the selected architectures.
2. Training and evaluating each architecture on:
 - original datasets
 - augmented datasets
3. Comparing performances across all experiments.
4. Analyzing the effect of augmentation on each architecture.
5. Producing reproducible experiments and detailed reports.

3. Datasets

The following datasets will be provided:

3.1. SHREC16

Triangular mesh classification dataset.

Provided versions:

- SHREC16 original
- SHREC16 augmented

3.2. ModelNet10

Triangular mesh version of ModelNet10.

Provided versions:

- ModelNet10 original
- ModelNet10 augmented

4. Architectures to Study

The official GitHub implementations must be used whenever available.

Architecture	Representation
DGNet	Mesh
TriangleMesh	Mesh
HalfedgeCNN	Mesh
MeshNet	Mesh
MeshCNN	Mesh
PointNet	Point Cloud

5. Important Constraints

5.1. Official Implementations

Official repositories must be used whenever possible.

The following information must be documented:

- repository link
- commit/version used
- modifications made

5.2. Dataset Consistency

All experiments must use the same train/test split for fair comparison

5.3. Reproducibility

The following elements must be documented:

- random seeds
- hyperparameters
- configuration files
- trained weights/checkpoints

5.4. Internal Augmentation Investigation

Each architecture must be investigated to determine whether:

- the original implementation already includes internal data augmentation
- online augmentation is applied during training
- preprocessing augmentation is integrated in the official pipeline

If internal augmentation exists:

- it must be disabled whenever possible
- all modifications must be clearly documented

The objective is to ensure fair comparison between original datasets and externally augmented datasets without interference from hidden augmentation mechanisms inside the architectures.

6. Experimental Protocol

Each architecture must be trained on:

7. Architectures to Study

The official GitHub implementations must be used whenever available.

Architecture	Representation	Official Repository
DGNet	Mesh	https://github.com/li-xl/DGNet
TriangleMesh	Mesh	https://github.com/EkanshGupta/TriangleMesh
HalfedgeCNN	Mesh	https://github.com/IngmarLudwig/HalfedgeCNN
MeshNet	Mesh	https://github.com/iMoonLab/MeshNet
MeshCNN	Mesh	https://github.com/ranahanocka/MeshCNN
PointNet	Point Cloud	https://github.com/charlesq34/pointnet

Total:

$$6 \text{ architectures} \times 4 \text{ dataset versions} = 24 \text{ experiments}$$

8. Required Evaluation Metrics

For each experiment, the following metrics must be collected.

8.1. Classification Metrics

Metric

Accuracy

Precision

Recall

F1-score

8.2. Computational Metrics

Metric

Total Training Time

Average Epoch Time

Inference Time

Number of Parameters

GPU Memory Usage

9. Training Information to Record

For every experiment, the following information must be provided:

Information
Number of epochs
Batch size
Optimizer
Learning rate
Scheduler
Loss function
Hardware used
GPU type
Training duration
Final convergence epoch

10. Required Visualizations

The following plots and visualizations must be generated for every experiment.

10.1. Training Curves

Plot
Training Accuracy
Validation Accuracy
Training Loss
Validation Loss

10.2. Comparative Visualizations

The final report must include:

- comparison tables
- plots
- augmentation impact analysis
- architecture comparison charts
- slot graphs

Examples:

- Accuracy comparison between models
- Accuracy improvement due to augmentation
- Training time comparison
- F1-score comparison